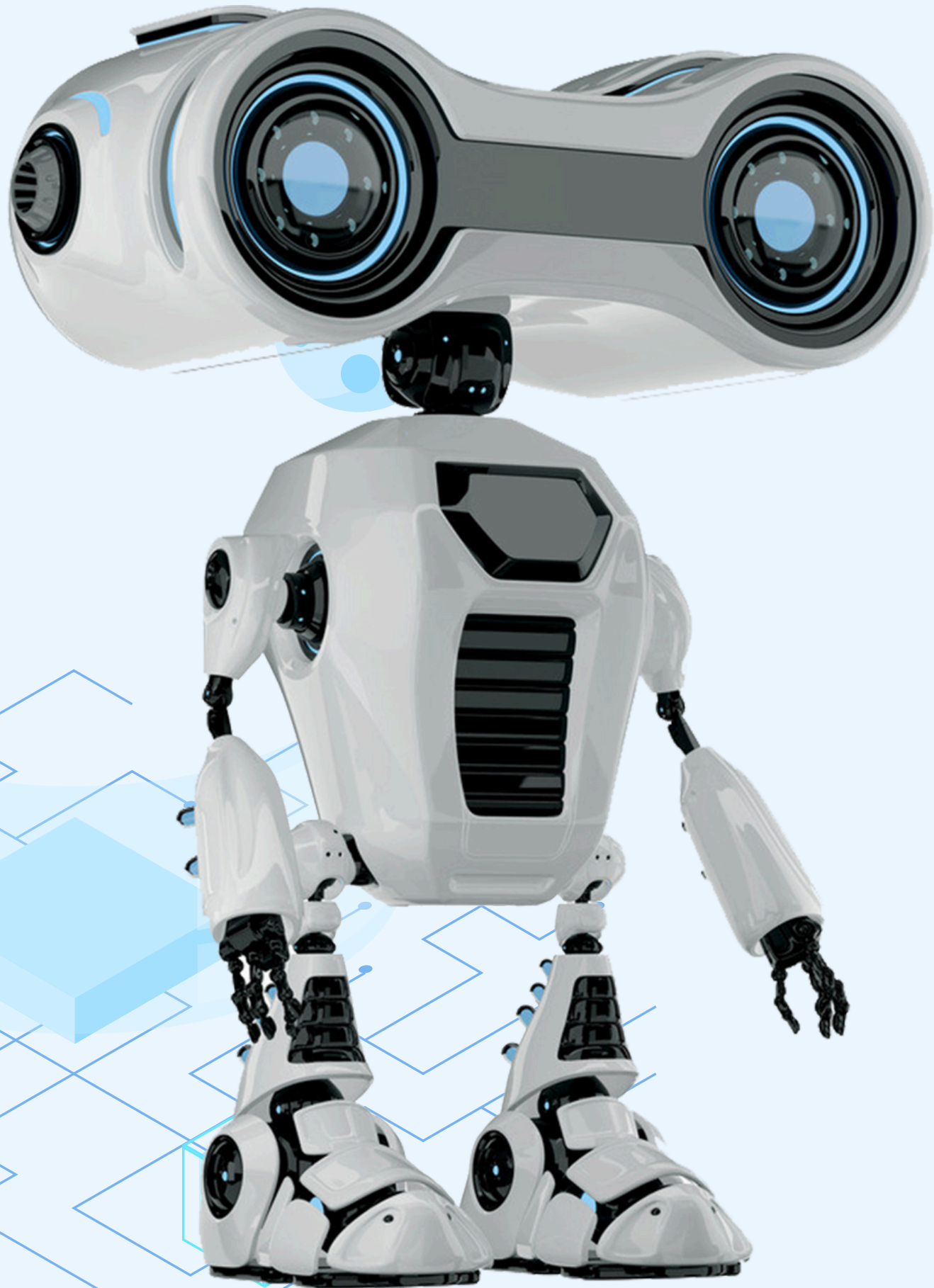




vFork



DataCrunch

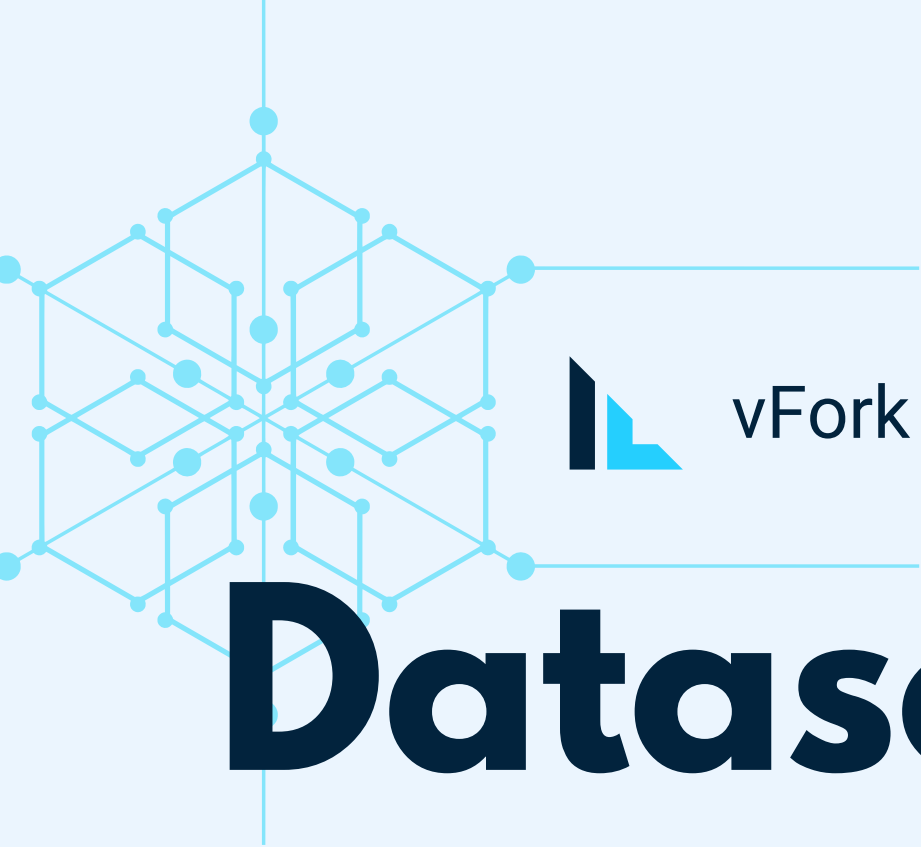
AgroChill



Problem Statement

The kingdom of Agrovia has multiple regions, each having an economic center that operates weekly. If crops couldn't be sold at the weekly market, they had to be resold at a loss. With the introduction of cold storage, a possibility opened up to use historical data to decide whether to sell crops fresh or to freeze them to sell at future markets for a better price. The goal of the forecasting model is to forecast accurate price predictions when given data about historical price variations.

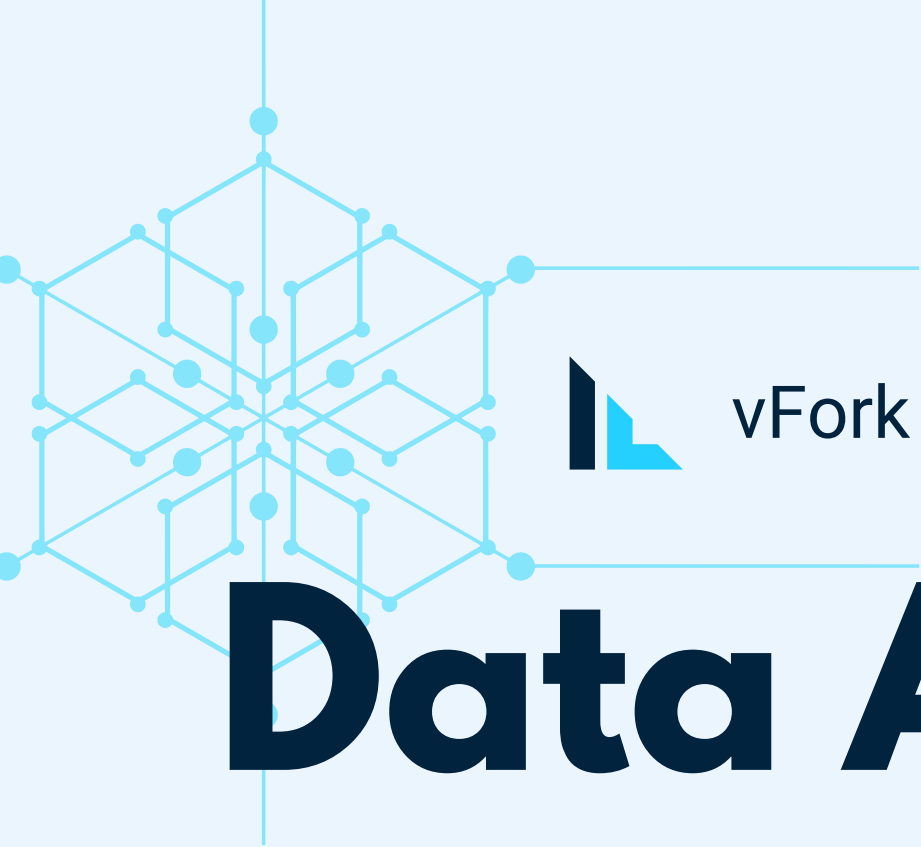




Dataset

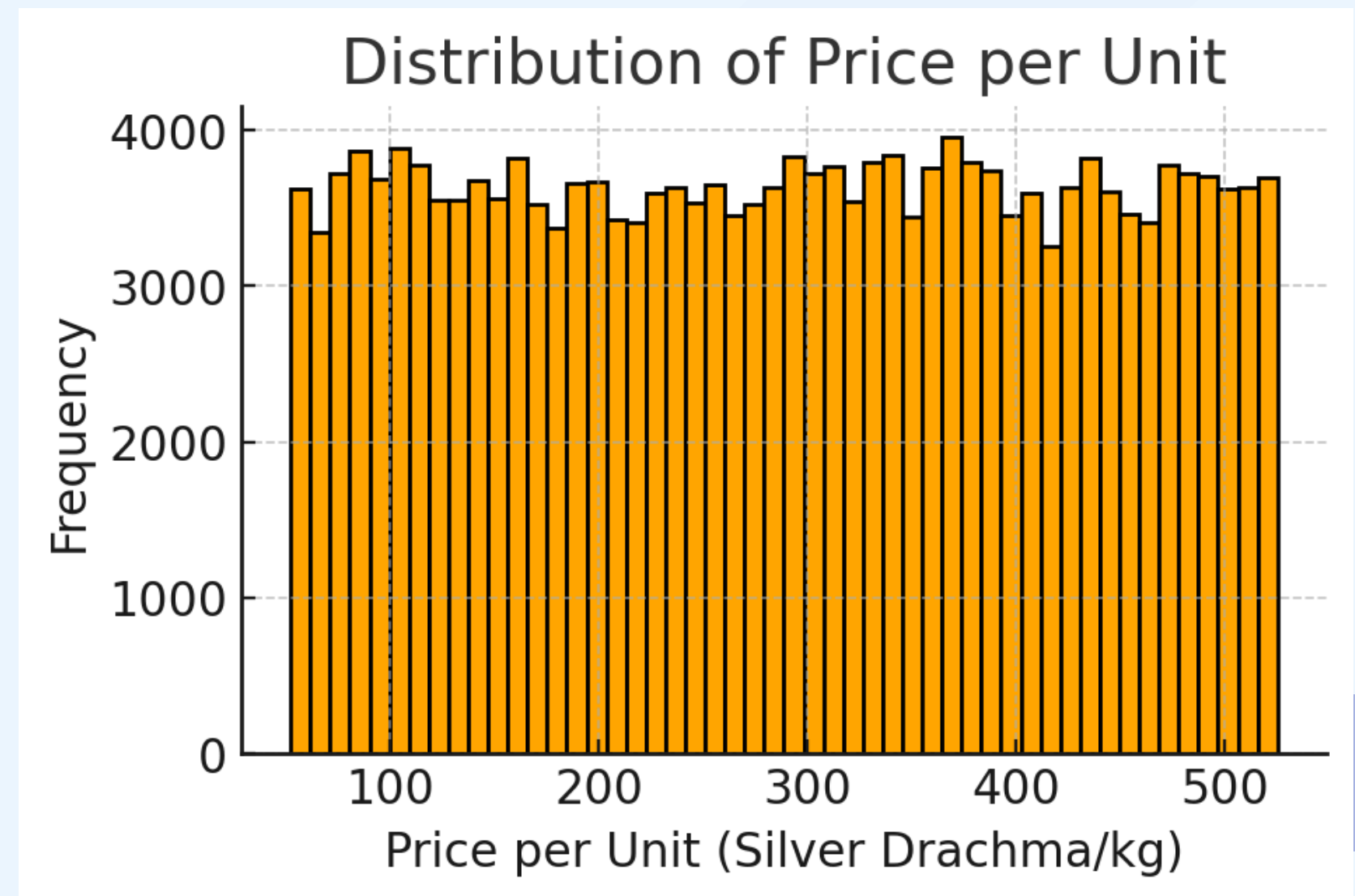
There are four datasets and two categories of data, weather data and price data. Weather data mainly consists of region, rainfall, temperature, humidity, and crop yield. Price data mainly consists of crop, region, and price.

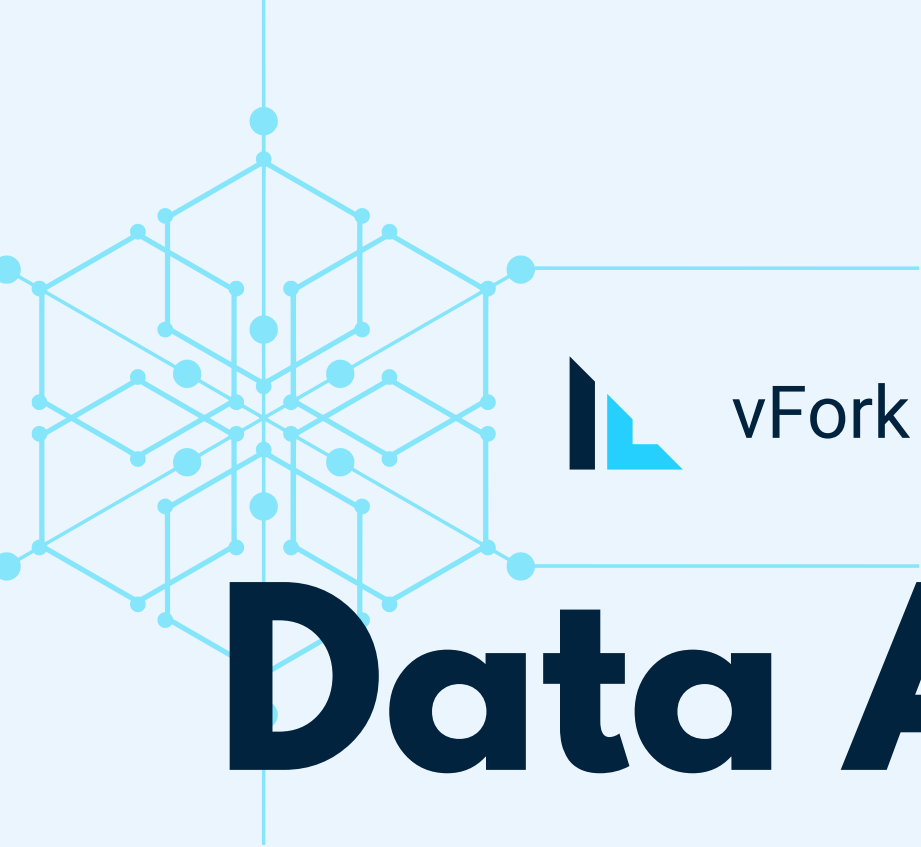




Data Analysis

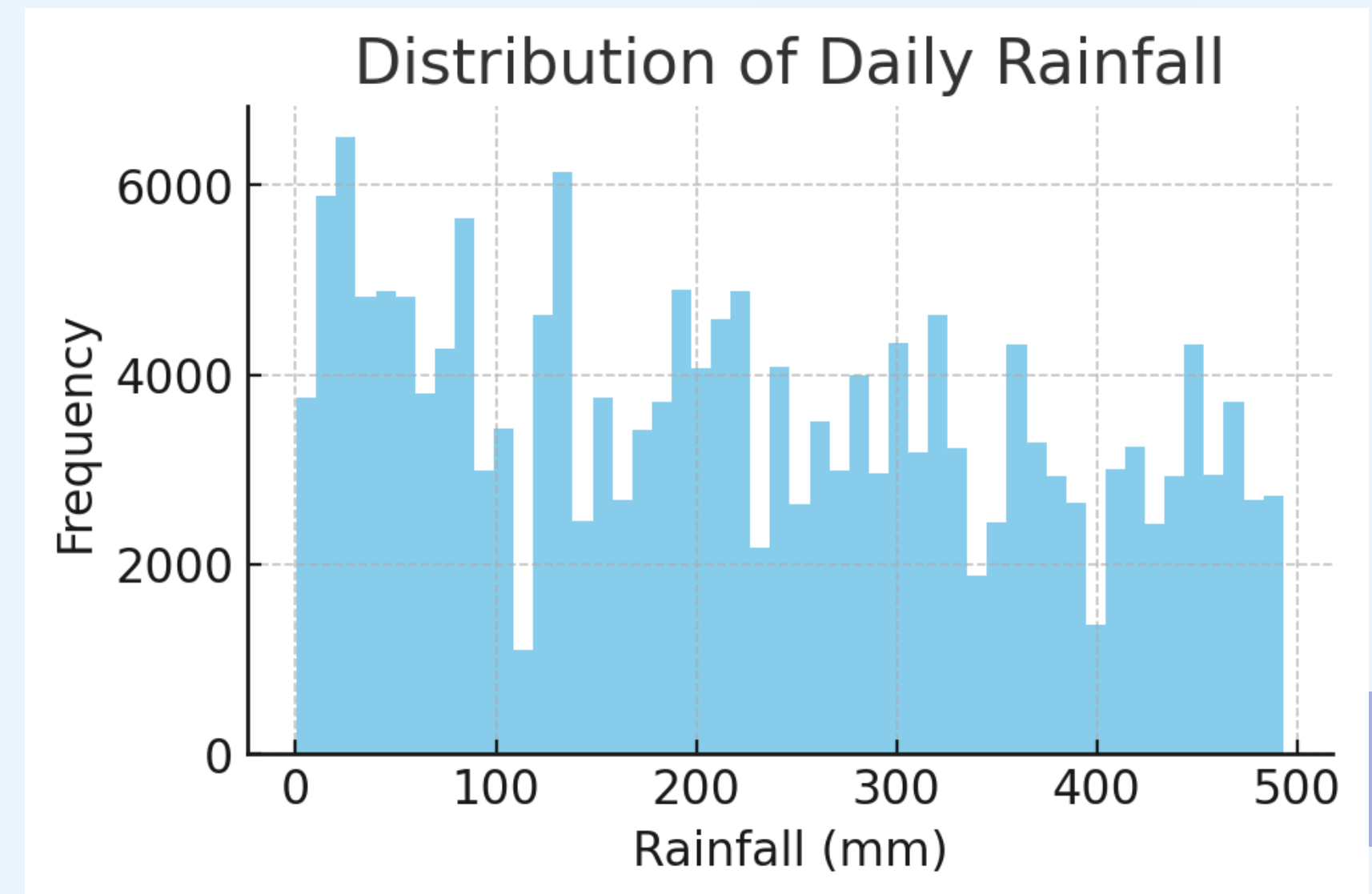
Distribution of individual commodity prices. Prices range roughly 50–525 Drachma/kg, with a broad, fairly even distribution and a slight central tendency around 250–300





Data Analysis

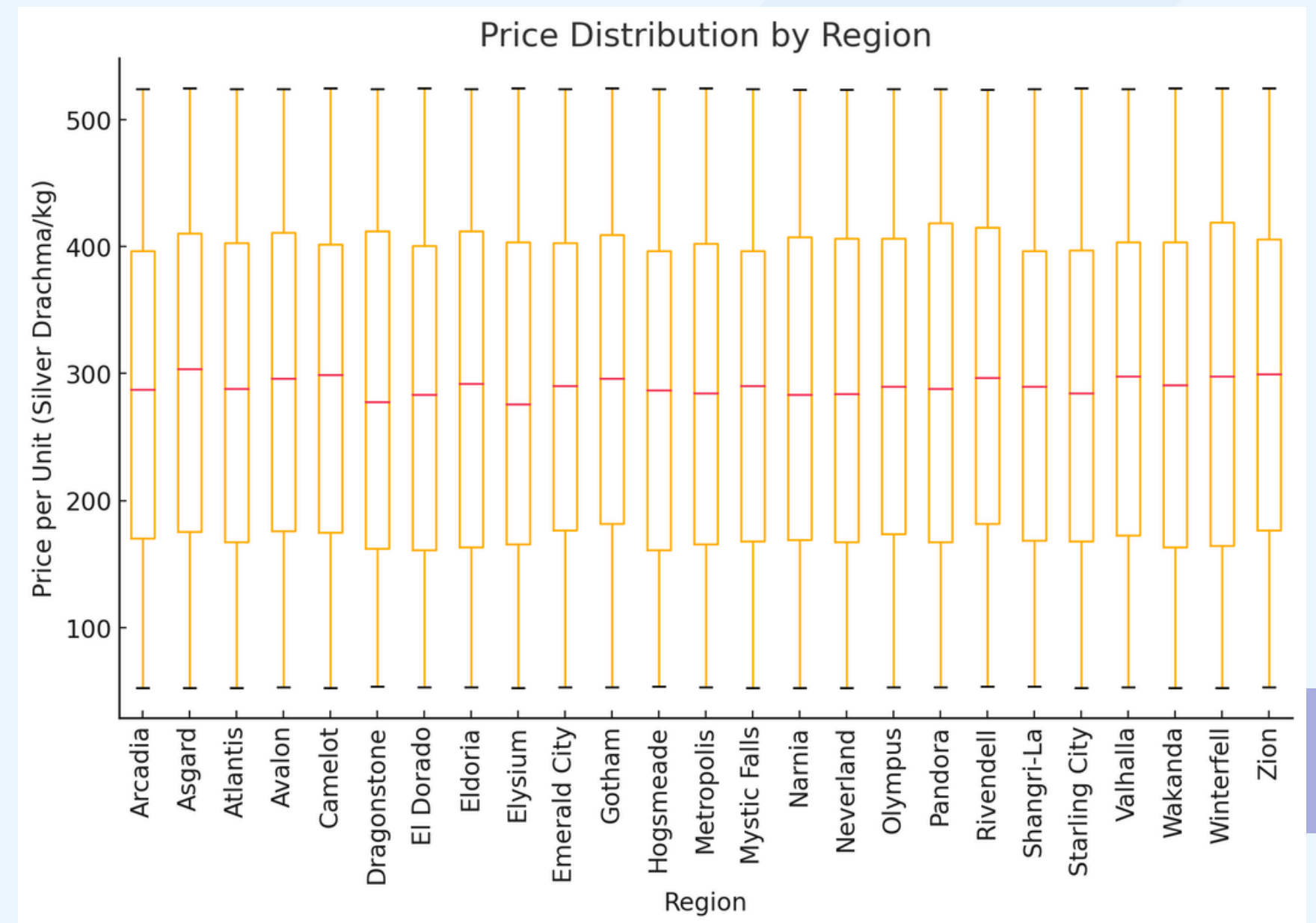
Distribution of daily rainfall. Most days see substantial rain (50–300 mm common), and there are outliers of extreme rainfall approaching ~500 mm





Data Analysis

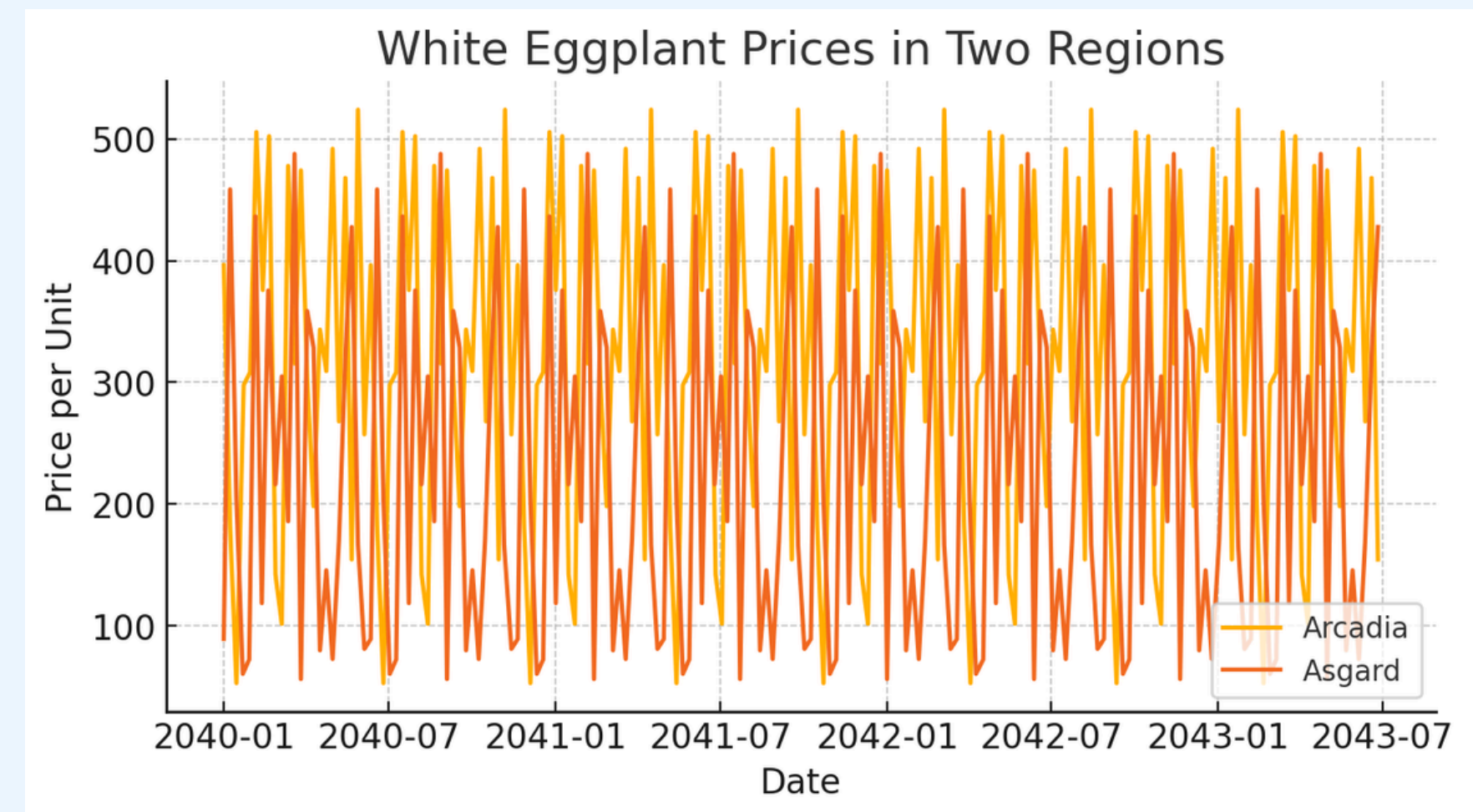
Distribution of prices by region (boxplots). Each region's prices (across all commodities and times) show a wide internal range (~50 to ~525) but very similar medians (red line) around 285–295. Regions like Gotham (far right) have a slightly higher median price, while El Dorado (toward left) is slightly lower, but the overlap is large





Data Analysis

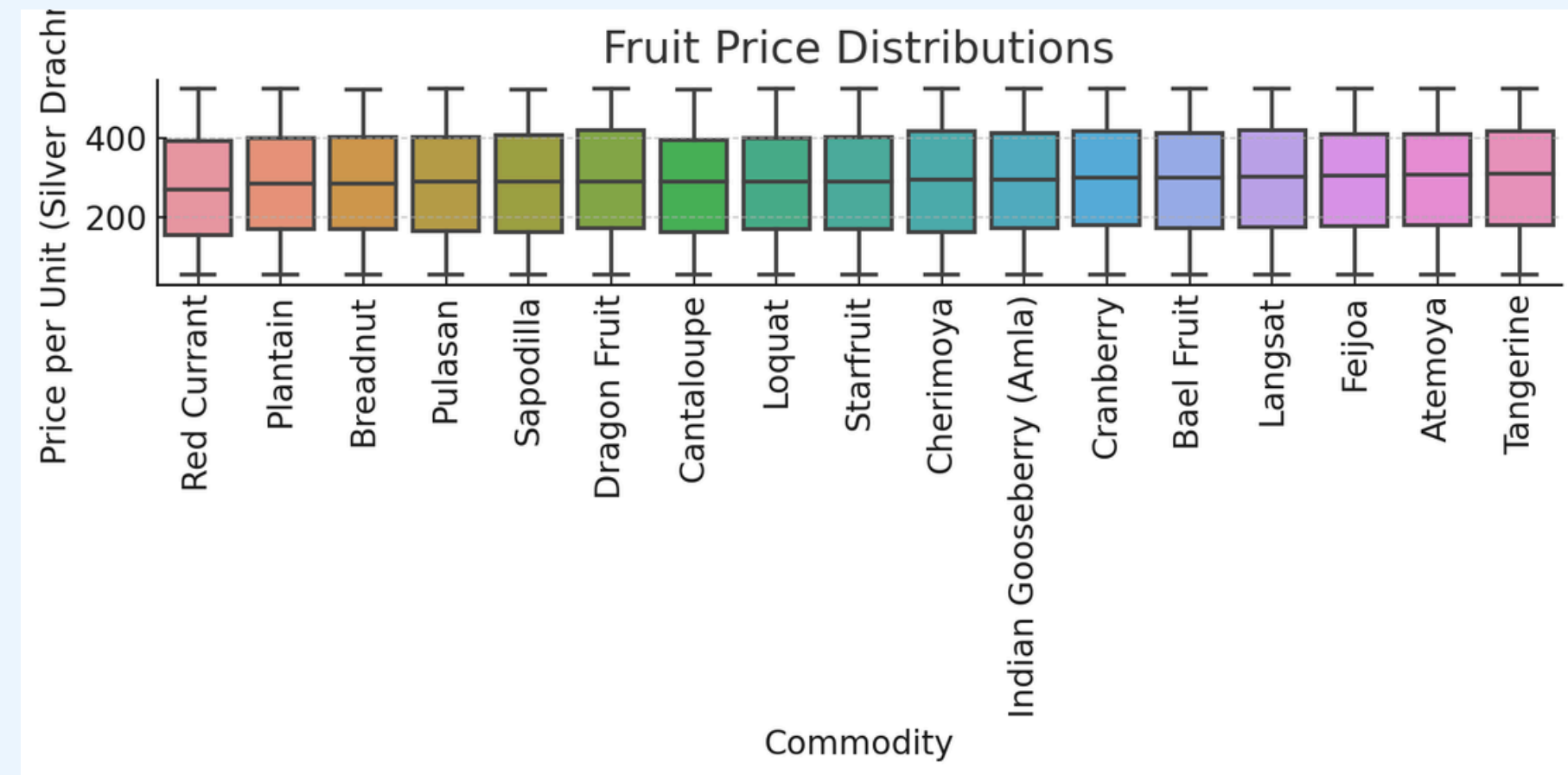
Weekly White Eggplant prices in two regions (Arcadia in orange, Asgard in yellow). The series is noisy week-to-week, but both regions show synchronized peaks and troughs. There is some repetition each year: for instance, prices dipped in early 2041 and 2043 (around 250 Drachma) and peaked in mid-2042 (around 520 Drachma)

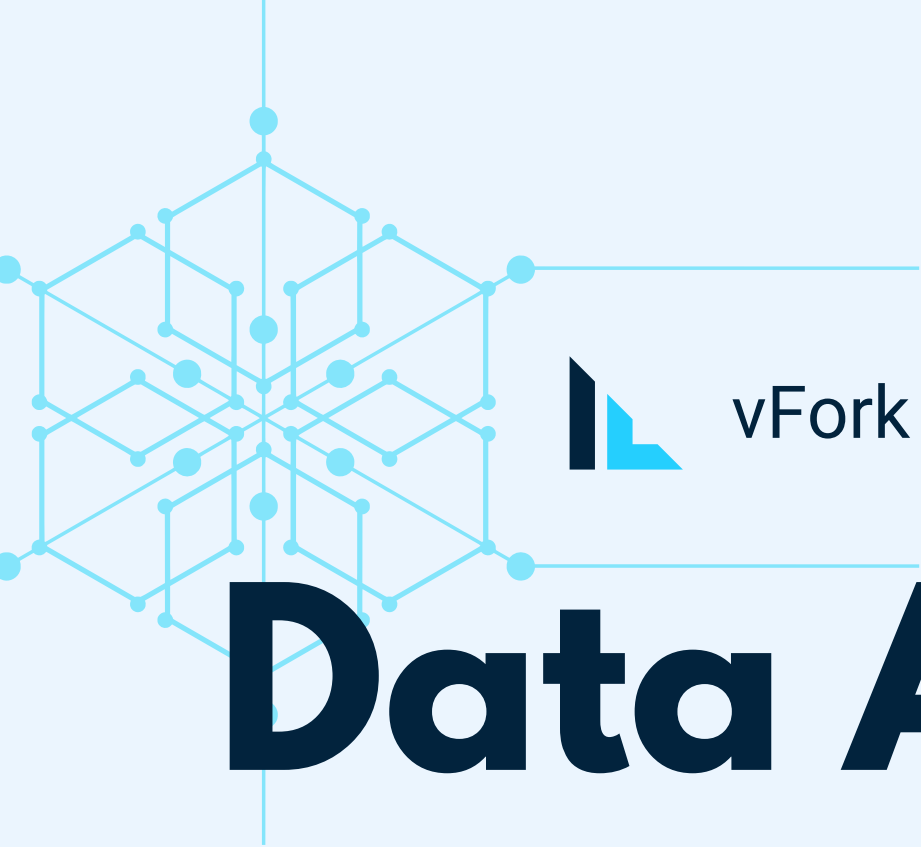




Data Analysis

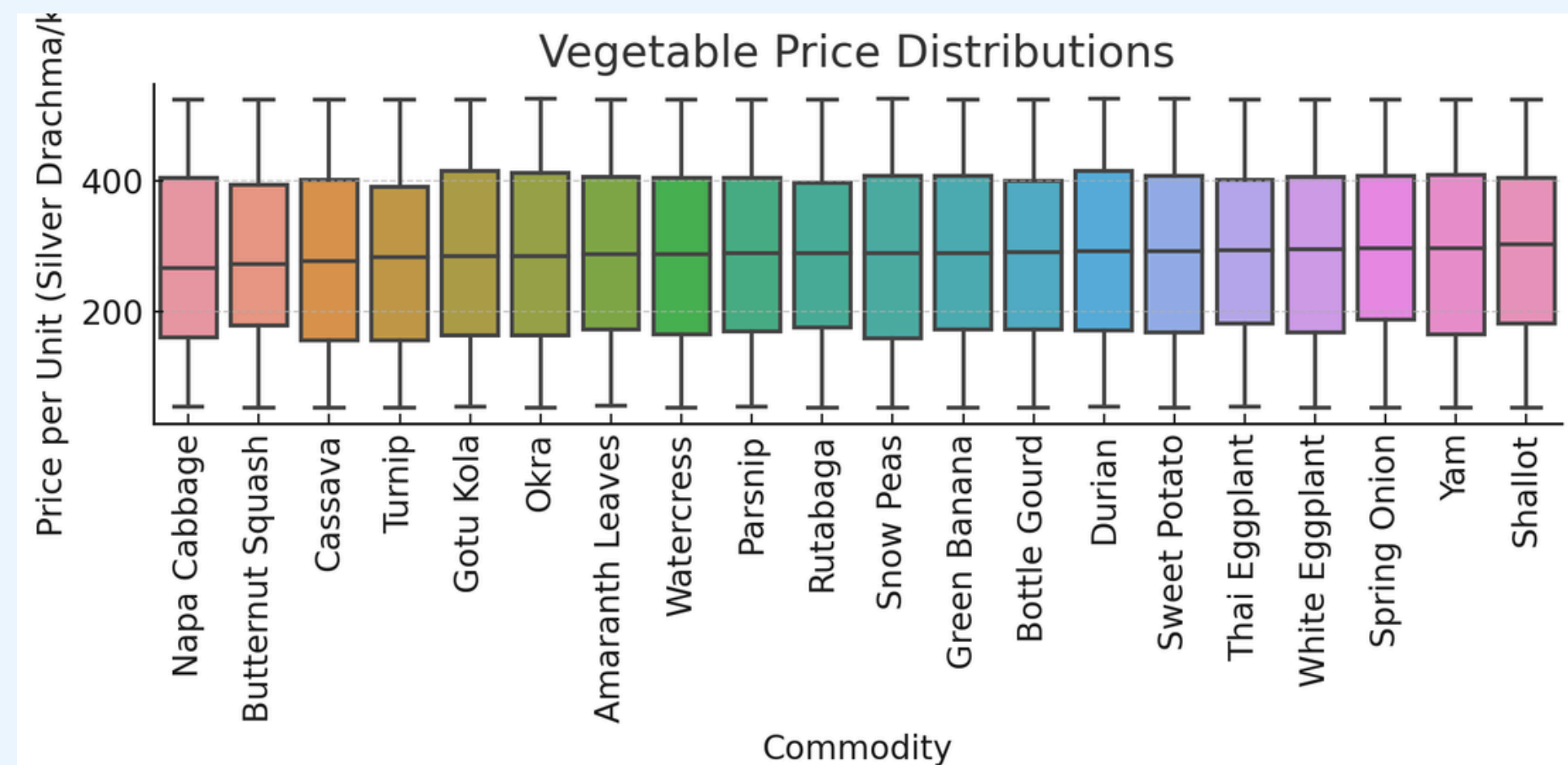
Price distributions for fruit commodities. The boxes show the median (line) and IQR for each fruit. All fruit medians lie roughly between 250 and 300. Some fruits like Red Currant and Plantain tend toward the lower end, whereas Feijoa, Atemoya, Tangerine are toward the higher end. Whiskers (and occasional outliers beyond them) illustrate that each fruit can reach low and high price extremes under certain conditions





Data Analysis

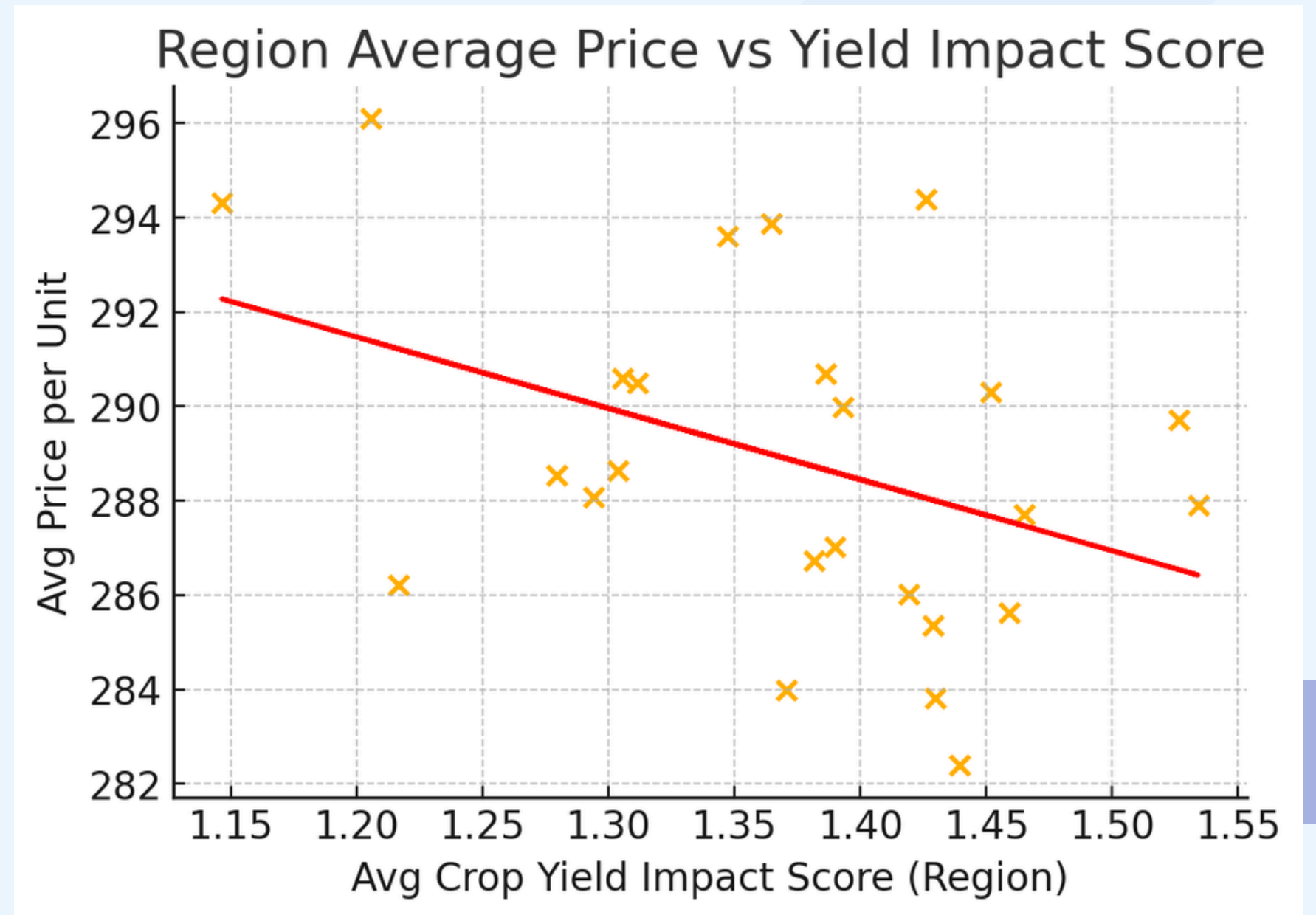
Price distributions for vegetable commodities. Vegetables similarly have medians in the 250–300 range. Napa Cabbage and Cassava are among the lower median price veggies, while Yam, Shallot, Spring Onion skew a bit higher. The spread of each box is large, indicating high volatility for all commodities





Data Analysis

Scatter of regions (each point is a region) showing average Crop Yield Impact Score vs average Price. There is a noticeable downward trend (red line): regions with higher yield score (better growing conditions) tend to have slightly lower prices on average





Features

- Since markets operate weekly, we performed weekly data aggregation, where the daily data provided in the datasets is aggregated into weeks. This created a weekly price series for all the region-commodity combinations.
- Additionally, we have added **lag features** to capture temporal dependencies in data (4 lag features to reflect 4 weeks ahead predictions).
- We added calendar features such as week, month, quarter, and year to allow the model to learn seasonal patterns.
- Categorical data, such as region, was one-hot encoded.

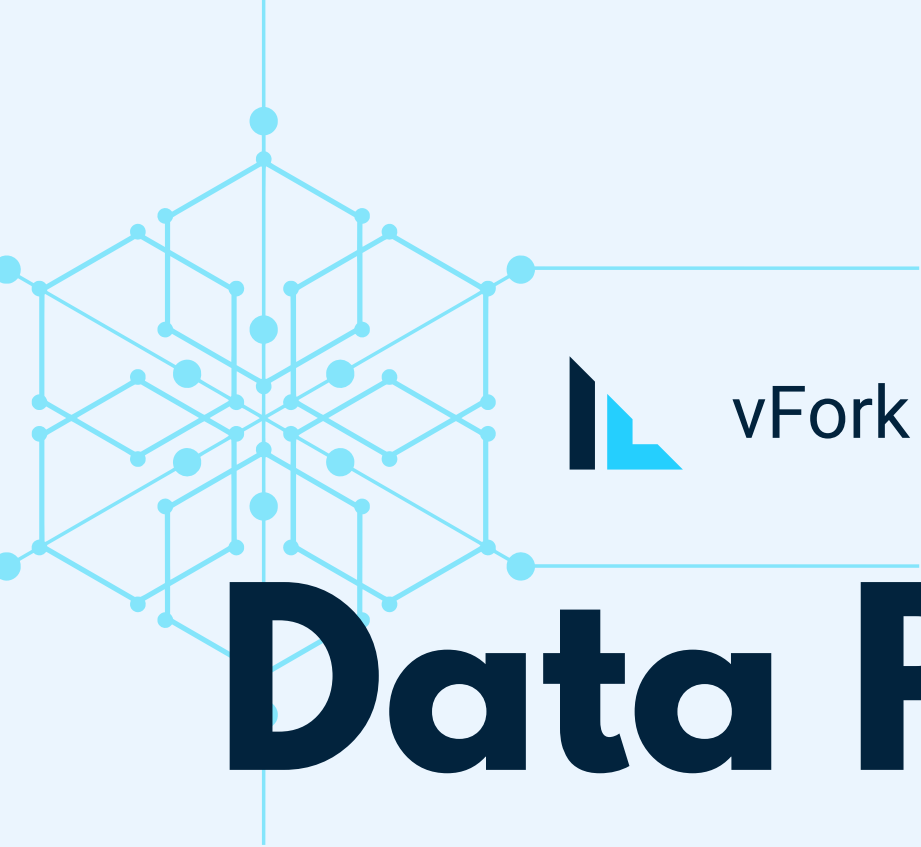




Model Selection

- Selected Model: **XGBoost**
- XGBoost is a light-weight tree-based model that can learn non-linear relationships.
- We trained a single XGBoost model per commodity and saved the model in JSON format.
- XGBoost is capable of online/incremental learning that is not facilitated in additive forecasting models such as Prophet.

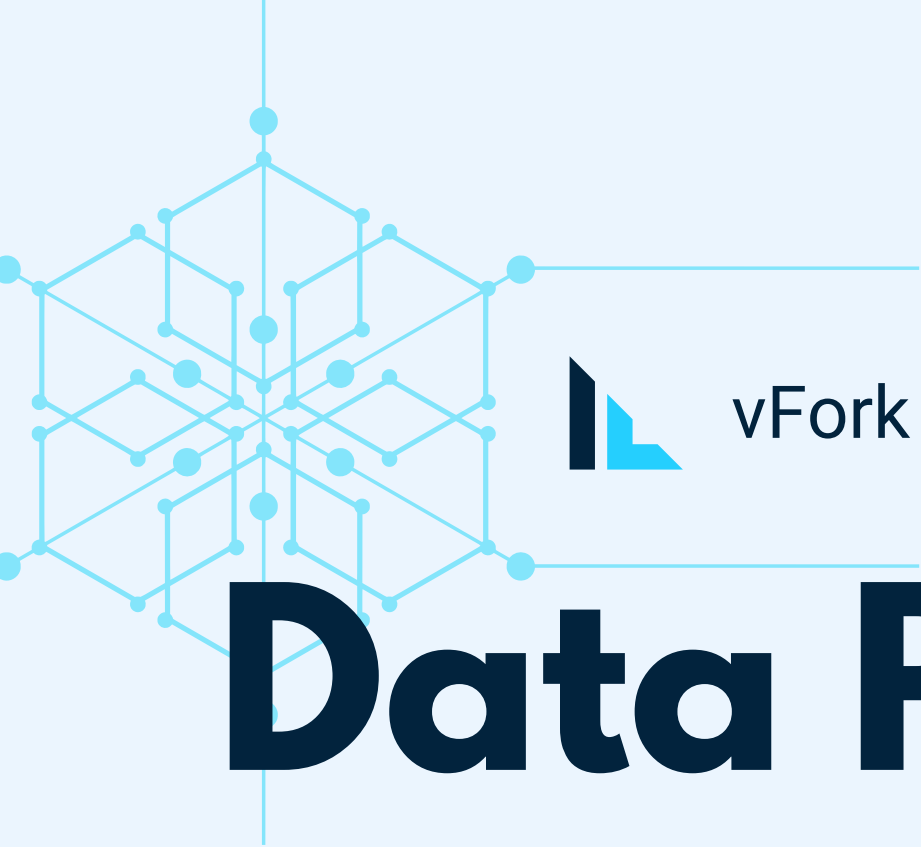




Data Pipeline

- We created a data pipeline that can ingest new data even after training the models.
- After training models for each commodity, the API endpoints can be used to ingest new weather or price data.
- New weather data affects all the models, while new price data only affects the models of the relevant commodities.





Data Pipeline Cont.

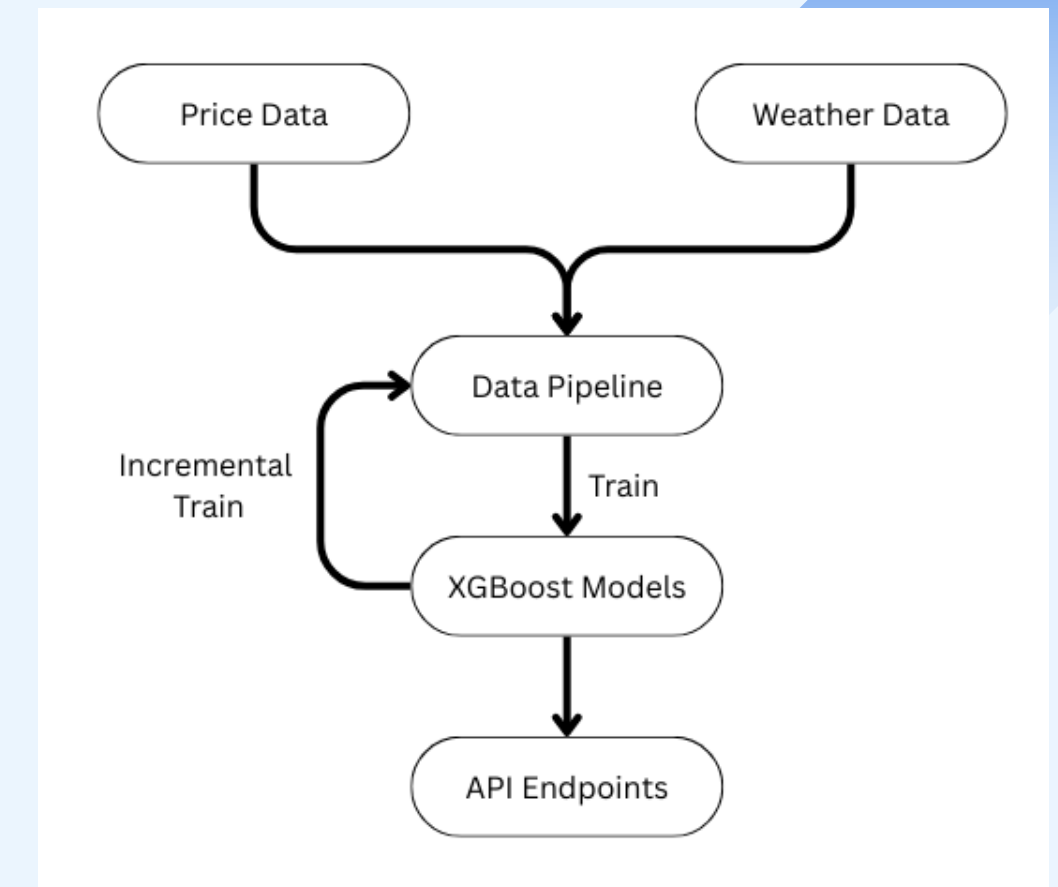
- The models are trained online/incrementally as new data is ingested.
- However, we didn't continue to do online training because, if the new data contains a semantic drift, then the model might catastrophically forget old data.
- Hence, we check the RMSE after each online training step and retrain all or specific models from scratch, if the RMSE rises above a threshold value.
- This threshold value is decided based on trial and error.





System Design

- The system is separated into multiple modules.
 - DataPipeline Class
 - The features module contains all the feature engineering functions
 - The model_training module contains all the training functions
 - The model_inference module contains all the functions related to inference
 - The model_evaluation module contains all the functions needed to evaluate all or a single module.
 - And finally, the main module contains the driver code, including all the endpoint functions.





Business Insights

- Optimal Sell/Store Decision
 - The model predicts 4 weeks ahead to the future.
 - If the model predicts high price values, then sell the crops.
 - If the model predicts low or normal price values, then freeze crops.
- Find the best market
 - The best market to sell might not be the nearest market.
 - With the newly built roads, crops can be sold in different regions.
 - The model can give predictions about which region might offer the optimal price in the upcoming weeks.





Business Insights Cont.

- Capacity Management
 - Since the market prices can be forecasted, you don't need to worry about overloading freezers with crops. If the market prices crash in the future, Magnus can decide when to rent freezers.
- Risk Management
 - Weather data can be used to identify seasonal natural disasters and can be used to decide whether to grow crops or not.





THANK YOU!